

Does chain-of-thought still protect against J-space ablation when problems get hard?

CS 2881 Homework 0 · August 3, 2026 · code: github.com/pgrindehollevik-harvard/cs2881

1 Hypothesis (pre-registered before any experiment; tag prereg-v1)

Gurnee et al. (2026) show that a model’s “J-space” can be deleted: at every token position, across a band of middle layers, one zeroes the residual stream’s projection onto the $k=10$ most active Jacobian-lens directions. In their frontier models this deletion destroys internal multi-step reasoning, spares shallow skills, and, most strikingly, spares GSM8K *when the model writes its reasoning out*. Written chain-of-thought appears to substitute for the internal workspace. We tested whether that protection survives on a much smaller model, Qwen/Qwen3-4B, and whether it survives increasing difficulty. **H1** (replication): on GSM8K, chain-of-thought protects more than direct answering does, beyond what matched random damage explains. Formally, the random-differenced protection $P_D = [R(\text{CoT}, J) - R(\text{dir}, J)] - [R(\text{CoT}, \text{rand}) - R(\text{dir}, \text{rand})]$ is positive (predicted magnitude $\geq +0.15$), where R is accuracy retention (ablated/clean). **H2** (extension): the protection *erodes* as problems get harder (GSM8K $\rightarrow$ MATH-500 $\rightarrow$ AIME; within MATH-500, levels 1 $\rightarrow$ 5). Rationale: externalization has limited bandwidth, and each written step of a harder problem still demands more internal workspace. Falsification criteria, sample sizes, floor rules, and a hypothesis-blind calibration gate were frozen in docs/PREREGISTRATION.md before any run. Every later departure is timestamped in docs/DEVIATIONS.md.

2 Experiment design

The design is a grid: two prompting modes (chain-of-thought vs. direct answer, same sampling settings) $\times$ three conditions (clean, J-space ablation, control ablation) $\times$ three datasets (GSM8K $n=150$; MATH-500 stratified 30 per level; AIME 2024+2025, all 60 problems). The control ablation is the design’s backbone: deleting ten directions damages any network, so at every (position, layer) we instead remove an *exactly equal-norm* component along random directions. Whatever survives the P_D subtraction is therefore specific to the J-space, not generic damage. A stop-gate stands in front of the main grid. Before it may run, the ablation must reproduce, on this model, both signatures the paper found: (1) a positive-control reasoning drop that is significant (exact McNemar) and at least $2\times$ the matched-random drop, with output degeneration below 10%; and (2) selectivity, meaning ordinary next-token prediction on held-out wikitext survives (top-1 agreement with the clean model above 0.80; our pre-registered quantification of the paper’s selectivity finding, whose exact values appear only in its Fig. 22 graphic). Ablation strength (the layer band and k) is calibrated only against these gauges. The calibration never sees a CoT-vs-direct comparison, so it cannot be tuned toward the hypothesis.

3 Experimental details

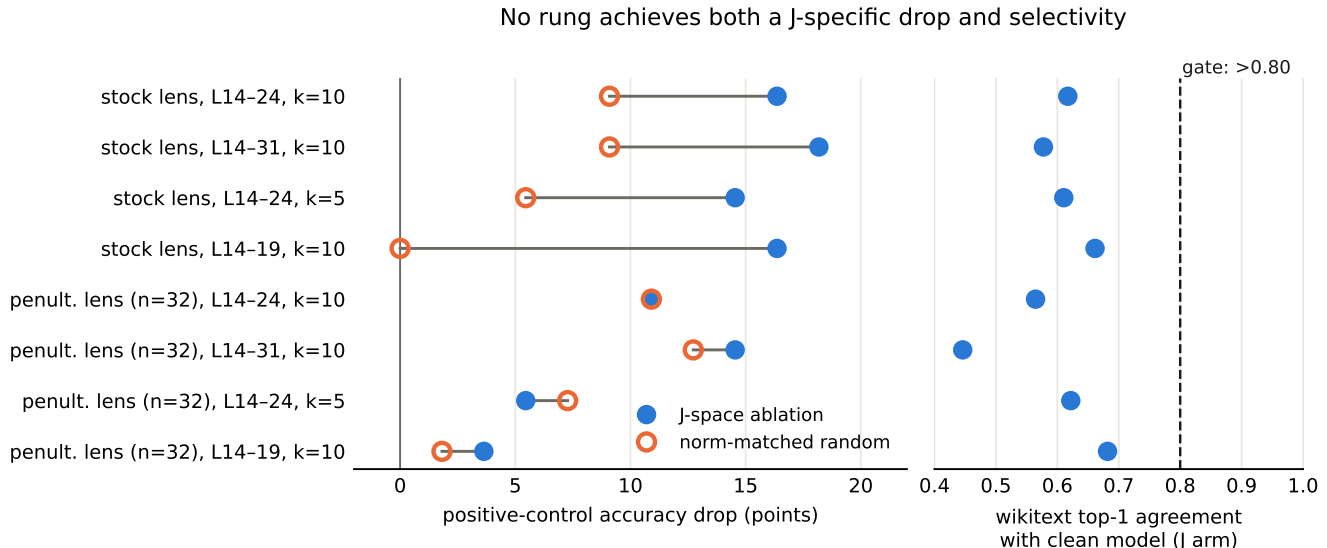
The companion repository provides the lens but no ablation code, so we implemented the intervention as forward hooks. At each (position, band layer): compute readout logits $\text{lm_head}(\text{rmsnorm}(J_\ell h))$; take the top-10 token ids, recomputed independently per position and layer; exempt any id in the clean model’s top-10 next-token predictions (a parallel clean stream over the same tokens supplies these); form directions $v_t = J_\ell^\top(\gamma \odot W_U[t])$; remove their joint projection by a regularized batched solve. Ablation applies to the prompt and to every generated token. The lens is the pre-fitted `neuronpedia/jacobian-lens` artifact, fit on this exact checkpoint. We validated it before use: it beats a logit-lens baseline at ranking silent multi-hop intermediates (pass@5 0.41 vs 0.37; pass@1 0.20 vs 0.19), though not on arithmetic order-of-operations readouts at rank 1; the pre-registered gate (win on either eval) passed. The paper reports its workspace band as percent depth, so we transferred it by proportion: candidates {14–24, 14–31} of Qwen3-4B’s 36 blocks, with a light band 14–19. Compute was one M4 Pro (48 GB, MPS); ablated decode ran 3.4–7.2 tok/s across calibration rungs (median ≈ 6 ; the ablated pass runs two streams plus per-layer readouts). Generation is chunk-resumable with 64-token persistence, which mattered: macOS update preparation repeatedly SIGKILLED the process (90s lifetimes at the worst), costing wall-clock but, by construction, no data. Grading uses `math-verify` on the last `\boxed{\}`; no-answer, truncation, degeneration, and wrong-answer are tracked as separate classes. Seven unit tests certify the intervention: exact identity when disabled, projections verifiably removed, the exemption rule honored, per-slot norm matching, seeded determinism, and chunked equal to unchunked.

4 Experimental results

Calibration. The paper’s positive control (“multihop”: silent two-hop factual composition, e.g. legs of the animal that spins webs) proved unusable here: clean Qwen3-4B scores only 0.42 where frontier models are near ceiling. This is itself a small-model datum. The pre-registered fallback control (“order-ops”: arithmetic order-of-operations completions, clean 0.745) was used instead. All four rungs then *failed* the stop-gate, so the main grid did not run on the stock lens:

Rung	J drop (pts)	random drop	wikitext top-1 (J / rand)	gate
14–24, $k=10$	−16.4 ($p=.004$)	−9.1	0.62 / 0.79	fail
14–31, $k=10$	−18.2	−9.1	0.58 / 0.73	fail
14–24, $k=5$	−14.5	−5.5	0.61 / 0.82	fail
14–19, $k=10$ (light)	−16.4 ($p=.012$)	0.0	0.66 / 0.84	fail

Read the table as two separate findings. First, the J-directions are causally real on Qwen3-4B. In the light band, deleting them costs 16 points on the control while exactly norm-matched random deletion costs *nothing* ($n=55$ paired items, exact McNemar $p=.012$), and outputs stay coherent (degeneration 3%). Second, the directions are not *selective*: 34–42% of ordinary next-token predictions change, whereas the paper finds ablation leaves “the bulk of ordinary text prediction intact” (§3.5.2). The damage always spills into everyday prediction. The pre-registered contingency then applied: the stock lens targets the final layer, while the paper prefers a penultimate target to avoid last-block artifacts, so we refit a penultimate-target lens and re-ran the ladder. Fit size was $n=32$ prompts rather than the registered $n\approx 100$, because 17 hours of MPS backward passes made $n=100$ infeasible (deviation 4, whose interpretation rule was committed before the re-run). All four rungs failed again, and diagnostically so: the refit lens’s “J-directions” behaved identically to the random control (drops 10.9 vs 10.9; 14.5 vs 12.7). Selectivity worsened at the $k=10$ rungs (0.45–0.56) and stayed comparable at the gentler rungs (0.62–0.68), never approaching the 0.80 gate. This is what estimation noise dominating a small fit should look like, which would in turn corroborate the stock lens’s J-vs-random gap as real structure; deviation 4’s registered confound means we treat that reading as suggestive rather than established.



5 Analysis of results

Verdict. H1 and H2 could not be answered on this model, and that is the finding. The pre-registered precondition for interpreting any CoT-vs-direct comparison (an ablation that is J-specific *and* spares ordinary prediction) was not achievable at any of eight settings across two lenses. We report the negative replication rather than an uninterpretable grid. **What is positively established:** (i) the Jacobian lens transfers to Qwen3-4B as a readout, beating the logit lens at surfacing silent multi-hop intermediates; (ii) its directions are causally load-bearing, since the light band shows a 16-point J drop against a zero norm-matched-random drop with coherent outputs; (iii) what fails to transfer is the paper’s selectivity signature, the separation of workspace from ordinary prediction. **Two readings remain.** (a) Small models route ordinary next-token prediction *through* their verbalizable representations, so the workspace/automatic separation is an emergent property of scale. This fits the paper’s own Haiku 4.5 warning and our finding that clean Qwen3-4B scores only 0.42 on two-hop silent composition. (b) Lens estimation quality at 4B (the stock lens early-stopped at $n=479$, with a final-layer target) smears workspace directions with general-prediction components. Our refit could not settle (b) at a feasible fit size, so it is reported as unresolved, not refuted. **Next with more compute:** run the same pre-registered ladder on Qwen3-14B and 32B, for which pre-fitted lenses exist. Reading (a) predicts the gate starts passing with scale; reading (b) predicts it keeps failing. Also: a full-budget penultimate lens, and thinking-mode traces. All code, prompts, seeds, the full 8-rung calibration records (including per-item outcomes), and the crash-tolerant pipeline are in the repository for exact reuse. The main grid’s records intentionally do not exist.